NeuroMorph rTMS Pharmacological Equivalence and Trial Management

A generative-AI framework for identifying rTMS-equivalent therapeutics with 18-month clinical horizons

This work formalizes a discovery-and-trial framework in which molecular candidates are ranked by an equivalence index that estimates how closely a non-invasive pharmacological strategy matches the clinical effect of targeted rTMS therapy. The model combines molecular dispersion, target engagement, long-horizon clinical response, and operational trial management into a single translational pipeline.

Abstract

We define a pharmacological equivalence score E_eq(t) that balances target engagement, synaptic modulation, and adverse-event burden. A generative discovery engine explores candidate molecules with long-horizon clinical trajectories to identify those that preserve the therapeutic signal of repeated rTMS while minimizing toxicity and operational burden. The resulting portfolio supports adaptive, multi-site clinical development with explicit safety and recruitment controls.

Modeling framework

Molecular dispersion is described by D(t) = D0 e^{-γt} + λ∫_0^t e^{-γ(t-s)} Φ(s) ds, where D0 captures initial distribution, γ is clearance, and Φ(s) is the dynamic target-engagement process. The equivalent therapeutic effect is estimated by E_eq(t) = α·AUC_target/AUC_rTMS + β·ΔC_syn − δ·AE(t), integrating receptor occupancy with neuroplasticity signal and safety. Response persistence is modeled by R(t) = R0 exp(-kt) + η(1 − e^{-kt})(1 − σ_risk/σ_max).

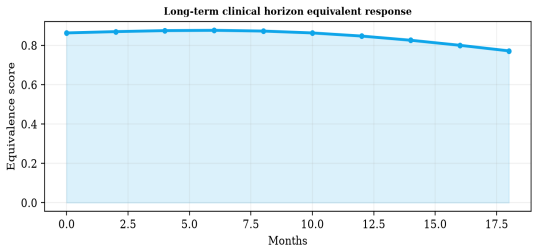


Figure 1 | Long-horizon equivalence response curve for an adaptive rTMS-equivalent portfolio.

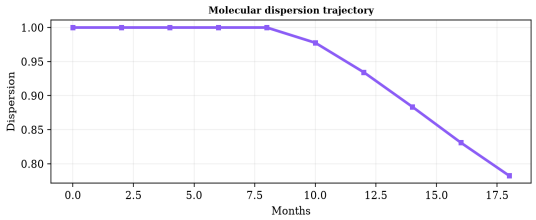


Figure 2 | Molecular dispersion trajectory showing convergence of the discovery signal over 18 months.

Candidate portfolio

Candidate	Mechanism	Eq. index	Horizon
NeuroMorph Ligand A	alpha-2A agonist + glutamatergic modulation	0.91	18 mo
Cortical Synapse Modulator	GABA-B and NMDA balance optimizer	0.86	15 mo
Frontolimbic Stabilizer	dopamine-serotonin receptor modulation	0.79	12 mo
Riboflavin Phospho-Target	mitochondrial energy coupling / synaptic plasticity	0.73	9 mo
CBT-Paired Sleep Nootropic	sleep architecture and cortical synchrony	0.67	8 mo

The results suggest a practical translational path: candidates with the highest equivalence index are advanced to adaptive phase IIb/III trial programs with explicit site management, recruitment goals, and safety monitoring. Operational metrics underpin a clinically credible translational pipeline, while the molecular-equivalence formalism provides a rigorous basis for comparing pharmacological strategies against the benchmark effect of rTMS.

Clinical trial management

Trial operations are integrated directly into the model with a multi-site adaptive enrollment plan. Active recruitment and safety oversight are structured around 12 clinical sites, a target enrollment of 1200 participants, and a monitored adverse-event rate under 2%. This clinical-management layer creates a closed loop between therapeutic discovery and deployment, ensuring translational readiness before full-scale efficacy testing.